Need Not Decide: A Benchmark for Judicial Non-Resolution in U.S. Federal Appellate Opinions

Anonymous ACL submission

Abstract

Deciding less than it could is one of the ways an appellate court exercises authority, and courts announce it in a small vocabulary that has been stable for a century. We ask whether language models can tell an issue a court resolved from one it discussed and left open, and what becomes of an unresolved issue as later courts cite it. A frozen thirteen-expression dictionary applied to 928,254 federal appellate opinions from the CourtListener bulk release finds explicit non-resolution language in 18.92%. From that population we draw a stratified benchmark of 306 annotated issues, pairing trigger-bearing passages with matched issues from trigger-free opinions and with two adversarial strata where the trigger is present but misleading. The dictionary scores well pooled, but that score is an artefact of anchor-centred sampling. Per stratum the systems separate: supervised models beat the dictionary on the stratum built to defeat it, and a fine-tuned encoder beats it outright on the court-held-out condition. Context helps to about ± 256 characters and then hurts. We release 79,974 judicially authored characterizations of prior decisions and use them to hand-verify cases where a later court calls a declined question a holding, which bears on how citation counts are read as evidence of what an opinion settled. The dictionary, benchmark, and pipeline are released.

1 Introduction

Courts are unusual among institutions in that their output is a written account of their own reasoning, so what they decline to do is recorded in the same medium as what they do. A United States appellate court that has raised a legal question is not obliged to answer it. It may resolve the case on a narrower ground and say that it "need not decide" the question. It may assume an answer "without deciding" and show that the outcome is the same either way. It may state that the question is left "for another day." These moves are ordinary, and they are marked in prose by a small vocabulary that has been stable for a century.

Declining to decide is a choice about how much law to make. It is the working form of the norm that a court should rule no more broadly than the case requires, and it is one of the few exercises of judicial discretion that leaves a direct textual trace instead of having to be inferred from votes or outcomes.

The practice matters for two reasons that pull in different directions. For courts, declining to decide is a discipline, the means by which a court keeps its holdings narrow and leaves a question open for a better-argued case. For anyone who later reads the opinion, it is a trap. An opinion that assumed a proposition without deciding it reads, at the level of the sentence, like an opinion that held it. The distinction lives in the surrounding paragraphs, and sometimes in a footnote several pages away.

That combination makes judicial issue disposition a natural test for language technology. The label depends on discourse structure rather than lexical content, the surface cue is easy to find and easy to misread, and there is an external record of what happened next. This paper builds a task around it and uses the task to ask an institutional question.

1.1 Two layers

We separate a classification problem from the phenomenon it lets us study. For an issue q appearing in opinion o, layer one asks for

$$D(q, o) \in \{\text{DECIDED}, \text{UNRESOLVED}\}, \quad (1)$$

with UNRESOLVED refined into AVOIDED, ASSUMED, and RESERVED. The primary benchmark is deliberately binary. Layer two takes an issue q that opinion o_0 left unresolved, follows the chain $o_0 \to o_1 \to o_2 \to \cdots$ of later opinions that cite o_0

in a discussion of q, and asks what decisional status each later opinion ascribes to o_0 on that issue.

1.2 Population, benchmark, and chains

We treat these as three populations rather than one. Prevalence is estimated over every eligible opinion, with natural frequencies preserved. The benchmark is a stratified sample that deliberately oversamples the informative minority. The longitudinal analysis is restricted to issues whose originating opinions have had time to be cited. Keeping them separate means no single statistical question is forced onto a sample chosen for a different purpose.

1.3 Contributions

1. A population-scale measurement of explicit judicial non-resolution across 1,998,580 federal appellate opinions, the Supreme Court and all thirteen courts of appeals, with the eligibility funnel reported in full (§4, §5).
2. A benchmark of 306 annotated issues whose negative class is not the absence of a phrase. Trigger-bearing passages are paired with matched issues from opinions containing no dictionary expression and with two adversarial strata built so the dictionary is actively misleading (§6).
3. An evaluation with four held-out conditions showing that the dictionary's pooled score is a property of the sampling, that supervised systems beat it on the adversarial stratum and on court-held-out generalization, and that context past a few hundred characters degrades every system family. A within-annotator ablation shows the same shape for a human (§7).
4. A released set of 79,974 judicially authored characterizations of prior decisions, and a hand-verified case series in which 4 of 10 propositions a court declined to decide are later described as that court's holding (§8).
5. A released package containing the dictionary and its hash, the guideline with its one recorded revision, the benchmark with splits and stratum labels, the citation chains, and the pipeline that rebuilds all of it.

2 Related Work

2.1 Legal benchmarks and resources

Legal NLP has moved toward suites assembled with practitioner input, of which LegalBench (Guha et al., 2023) is the clearest example, alongside resources targeting specific reasoning steps or document types (Zheng et al., 2021; Chalkidis et al., 2022; Hendrycks et al., 2021) and recent workshop resources spanning jurisdictions (Xie et al., 2024; Jang et al., 2025). Our task differs in what supplies the label. CaseHOLD asks which holding belongs to a citation. We ask whether a passage states a holding at all, a question with no answer at the level of the citing sentence.

2.2 Precedent treatment and citators

The closest existing task is classifying how a later case treats an earlier one, the problem commercial citators address with editorial teams and colour-coded flags. Demir and Canbaz (2025) benchmark language models on multi-label precedent treatment over expert-annotated citations. That work asks whether a case is still good law, taking as given that the case decided something. We ask the prior question, which no citator records. A case can carry a clean flag while the proposition later opinions attribute to it is one the court expressly declined to decide. Treatment classification takes the case as its unit and we take the issue, so the two signals are close to orthogonal.

2.3 Context, stability, and what models use

Several recent results argue that apparent legal competence is sensitive to how evidence is positioned and how the question is posed. Xia et al. (2025) show that long-context recall on court opinions depends on surrounding material rather than on raw retrieval, Purushothama et al. (2025) find legal interpretation judgments unstable against human judgments, Held and Habernal (2025) report that extracting formal argument structure remains hard, and Blair-Stanek et al. (2024) show that even basic operations over legal text are unreliable. Two choices follow. We vary the context window rather than fixing it, because the evidence that settles our label is often paragraphs from the cue, and we report accuracy per stratum rather than pooled, because a pooled number over a constructed benchmark reports the construction. Domain pretraining (Chalkidis et al., 2020) and large open corpora (Henderson et al., 2022) have made appellate text a standard substrate, and CourtListener (Free Law Project, 2026) supplies the bulk release we use. Work on citation networks in law has largely operated at the case level (Fowler et al., 2007; Black and Spriggs, 2013). We build chains at the level of an individual issue.

2.4 Hedging, uncertainty, and factuality

Detecting hedged assertions has a long history in scientific and biomedical text (Farkas et al., 2010; Vincze et al., 2008), and event factuality annotation formalizes the graded commitment a text makes to a proposition (Saurí and Pustejovsky, 2009; Rudinger et al., 2018). Judicial non-resolution is a domain-specific instance whose commitment is institutionally consequential and is contested by later readers whose characterizations we observe. Related in spirit, Hou et al. (2024) distinguish gaps from hallucinations in generated legal analysis.

2.5 Avoidance in legal scholarship

Legal scholarship has long studied constitutional avoidance and the narrowness of holdings, generally through close reading of small numbers of prominent cases. We do not adjudicate that literature. We supply the measurement layer it lacks, namely how often the practice occurs across an entire appellate system and what becomes of the issues it leaves behind.

3 Task Definition

3.1 Items

An item is a triple (o, a, q), an opinion o, an anchor span a locating a passage inside it, and a neutral statement q of the proposition that passage is about. Issue statements are generated from the text around the anchor and constrained to be uninformative about the answer. Any dictionary expression or holding cue appearing in a candidate statement is cut before the statement is kept, so no issue statement contains one. Appendix D gives the extraction procedure and its yield.

3.2 Layer one

The annotator sees q and a passage from o and assigns $D(q, o)$. DECIDED means the authoring court resolved the issue, whether or not the resolution was necessary to the judgment. UNRESOLVED means it did not resolve the issue for itself. Three rules do most of the work. The label describes what *this* opinion did, so a sentence reporting that the district court thought it need not decide something says nothing about this court. In a cluster with separate opinions each opinion is its own item, and a dissent that would decide an avoided issue is DECIDED. A court may write that it need not decide an issue and then decide it anyway, in an alternative holding or a footnote, and where it does the label is DECIDED. The full guideline is Appendix E.

3.3 Layer two

Given an issue q in origin o_0 and a passage in a later opinion o_1 that cites o_0, the annotator assigns a status level

$$\begin{array}{ll} L_0 & \text{explicitly unresolved} \\ L_1 & \text{assumed or conditional} \\ L_2 & \text{supported or probable} \\ L_3 & \text{treated as established} \end{array}$$

and two independent binary flags. $A = 1$ when o_1 attributes a resolution of q to o_0. $E = 1$ when o_1 resolves q itself, on its own analysis.

The distinction between A and E is the analytical core of layer two. If *Smith* said it need not decide X and *Jones* says “the issue remains open, but we now hold X,” that is doctrinal development rather than error, and $A = 0$, $E = 1$. If *Jones* says “*Smith* held X,” that is attributional escalation, recorded as $A = 1$. We use the term *precedential status escalation* for $\mathrm{Status}_{\text{later}}(q) > \mathrm{Status}_{\text{origin}}(q)$ and reserve any stronger word for cases the annotation establishes to be inconsistent with the origin.

4 Corpus and Population

4.1 Source

We use the CourtListener bulk release of 30 June 2026: the opinion, opinion-cluster and docket tables, the citation map, and the reporter-citation table. The export escapes embedded quotation marks with a backslash rather than by doubling them, so parsing with the CSV default desynchronizes the reader on roughly two thirds of cluster rows. We note this because the failure is invisible in the resulting file. All processing is a single streaming pass on one laptop.

4.2 Population

The population is every opinion issued by one of fourteen courts, the Supreme Court and the thirteen courts of appeals. CourtListener holds other federal appellate bodies, among them the historical circuit courts, the Court of Customs and Patent Appeals, and several agency review boards. These are outside the population and Appendix A lists every excluded court. Because the cluster table carries a docket key but no court, the docket table is the only route from an opinion to its issuing court.

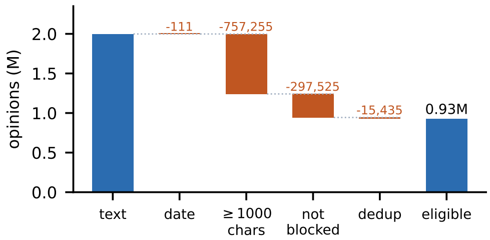

Figure 1: Eligibility funnel. Each bar is the count surviving one criterion, applied in the order shown; the thousand-character minimum is the largest single reduction.

4.3 Eligibility

Criteria are applied in a fixed order and each one's cost is shown in Figure 1. We require a parseable filing date in $[1789, 2026]$ and at least one thousand characters of extractable text, which removes one-line orders, judgment entries, and the volume of Supreme Court orders-list records. We further require that the record not be blocked from public view, and we deduplicate, since the same opinion is often ingested from more than one upstream source.

4.4 Text

Opinions are stored in several markup flavours. We take the first non-empty field in a fixed precedence order and strip markup, recording which field supplied the text so that source composition can be reported by period. Appendix B gives the precedence and the composition.

5 Measuring Non-Resolution at Population Scale

5.1 The frozen dictionary

Candidate generation uses thirteen expressions, fixed before annotation and unchanged afterwards, listed in Table 3. Its SHA-1 is `c23210667a4e`. Patterns tolerate a few intervening words, so "need not now decide" is captured alongside the bare form. One expression is operationalized rather than matched literally. The design names the stem *reserve*, which in appellate prose is dominated by senses unrelated to non-resolution, so the pattern requires it to govern a word denoting a legal question or a deferral.

5.2 Lexical prevalence is candidate generation, not the measurement

A count of opinions containing these expressions is not an estimate of judicial avoidance. It is an upper bound on one observable form of it, inflated by passages describing other courts and deflated by courts that leave issues open in ordinary prose. We report it because it fixes the sampling frame and because it is a population census rather than a sample. For a stratum of N eligible opinions of which k contain at least one expression we report

$$\hat{\pi} = k/N \quad \text{with} \quad \hat{\pi}^{w} = \frac{k + z^2/2}{N + z^2}, \qquad (2)$$

the Wilson centre (Wilson, 1927), giving an interval even though the population is enumerated, because the expression is a noisy proxy for the practice rather than a sample from it. §7 and Appendix G quantify both errors.

5.3 Analysis hierarchy, fixed in advance

A corpus this size supports a great many regressions, so the order of questions was fixed before the benchmark was drawn. Three primary results: whether models separate DECIDED from UNRESOLVED, whether wider context helps, and what fraction of unresolved issues are later ascribed greater decisional force. Five secondary questions were pre-specified, covering court level, era, opinion type, precedential status, and citation volume. Everything else is exploratory and labelled as such.

5.4 What the population looks like

Figure 1 gives the eligibility funnel. Of 10,798,347 opinion records in the bulk table, 1,998,580 belong to the fourteen courts and carry extractable text, and 928,254 survive the eligibility criteria. The largest single reduction is the thousand-character minimum, which removes short orders and judgment entries; Supreme Court records account for a disproportionate share of these, because orders-list entries are stored as opinions. Deduplication removes a further tranche of records that are the same opinion ingested from more than one source.

5.5 How often courts say they are not deciding

175,610 eligible opinions, or 18.92%, contain at least one of the thirteen expressions, with 250,781 occurrences in total across 2.67 billion words. Figure 3 breaks this down by expression and Table 1 by court. The rate is not uniform, running from 12.27% in the 4 Cir. to 26.99% in the 1 Cir..

Figure 2 and Table 12 give the temporal picture. Both the opinion-level rate and the rate per hundred thousand words are reported, because opinion

court	N	k	%	95% CI	per 100k words
Supreme Court	51,064	8,983	17.59	17.26–17.92	7.0
1 Cir.	35,427	9,560	26.99	26.53–27.45	10.9
2 Cir.	79,681	15,779	19.80	19.53–20.08	10.1
3 Cir.	54,354	13,240	24.36	24.00–24.72	10.2
4 Cir.	80,435	9,869	12.27	12.04–12.50	8.7
5 Cir.	103,306	18,756	18.16	17.92–18.39	10.0
6 Cir.	73,904	11,851	16.04	15.77–16.30	7.2
7 Cir.	72,028	13,086	18.17	17.89–18.45	7.7
8 Cir.	79,280	12,756	16.09	15.84–16.35	8.4
9 Cir.	147,809	28,392	19.21	19.01–19.41	12.1
10 Cir.	45,445	9,207	20.26	19.89–20.63	9.6
11 Cir.	40,392	9,785	24.23	23.81–24.65	10.0
D.C. Cir.	43,022	10,020	23.29	22.89–23.69	9.3
Fed. Cir.	22,107	4,326	19.57	19.05–20.10	8.7

Table 1: Opinions containing at least one dictionary expression, by court. N is eligible opinions and k those with at least one expression. Intervals are Wilson (Wilson, 1927).

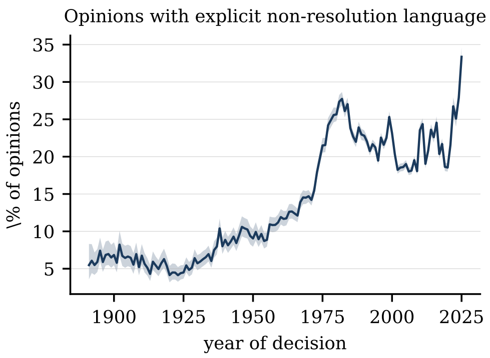

Figure 2: Share of eligible federal appellate opinions containing at least one frozen dictionary expression, by year of decision, with Wilson intervals. Years with fewer than 150 eligible opinions are omitted.

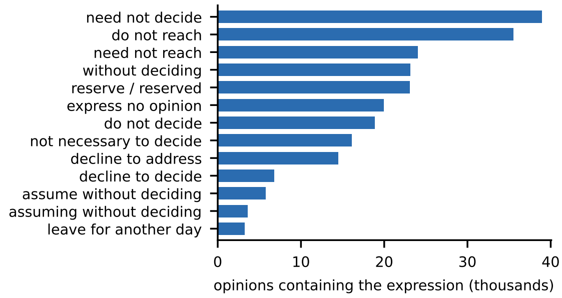

Figure 3: Opinions containing each of the thirteen frozen expressions. T08 and T09 are subsets of T07 by construction.

length is not constant across eras and an opinion-level rate alone confounds the two. The corpus itself changes over this span, in coverage, in digitization source, and in the share of unpublished dispositions, so the series should be read as a property of the available record rather than as a clean measurement of judicial behaviour over two centuries.

Opinion type and precedential status were named as secondary analyses before the corpus was built; Appendix H reports both.

6 Building the Benchmark

If the negative class is passages without a trigger, the task reduces to phrase recognition and any result is about the dictionary rather than about disposition. Three moves address this.

6.1 Matched controls

For each trigger-bearing opinion we draw a control opinion containing no dictionary expression anywhere, matched on court, period, opinion type, and length decile, and locate an issue inside it. Two locators are used: a holding cue (CTRL_HOLD) and a neutral marker naming an issue without indicating how it came out (CTRL_RAND), so the benchmark stands independent of the assumption that decided issues are the ones near "we hold."

6.2 Adversarial strata

Two structural flags identify passages where the dictionary is likely to mislead. H1 marks a trigger whose subject is plausibly another institution or that sits inside a quotation. H2 marks a trigger followed later in the same opinion by a holding sentence about the same proposition, the shape of a court that says it need not decide and then decides. Neither flag assigns a label; both are sampling strata, and annotation settles what the passage does.

6.3 Neutral issue statements

Each item carries a statement of the proposition at stake, constructed so it cannot leak the answer:

stratum	dec.	unres.	unclear	% unres.
CTRL_HOLD	72	0	1	0.0
CTRL_RAND	14	0	16	0.0
H1	10	57	31	85.1
H2	6	59	0	90.8
PLAIN	5	83	10	94.3

Table 2: Annotated labels by sampling stratum. The last column excludes UNCLEAR.

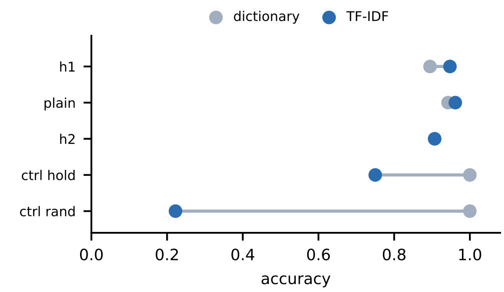

Figure 4: Accuracy by sampling stratum at the anchor sentence. The dictionary is at ceiling on the controls by construction; the informative comparison is H1.

any dictionary expression or holding cue is cut before the statement is kept. Appendix D gives the procedure and the 61.9% yield.

6.4 Sampling and splits

1,800 items were drawn with at most one per opinion, so context windows never overlap and splits are disjoint at the opinion level. Allocation across strata follows the pilot (Appendix F). Splits are nested and fixed in order: every item from 5 Cir., D.C. Cir. forms the court-held-out condition, then every remaining item filed in 2018 or later forms the temporal condition, and what is left is split at random into test and development. Only the development portion was used to fit anything.

6.5 Annotation

364 items were annotated under the frozen guideline, 58 of them UNCLEAR, leaving 306 with a usable binary label. Items were presented in a fixed random order ignoring stratum, split, court, and period, so the annotated set is a random subsample of what was drawn. Table 2 gives labels by stratum.

7 Can Models Tell?

7.1 Systems

Five are compared at four context widths: the frozen dictionary as a floor; TF-IDF with logistic regression and two fine-tuned encoders, supervised on the 120-item development split; a zero-shot open model, Phi-3.5-mini-instruct, scored by the likelihood of the two label continuations so it never generates; and a classifier seeing only the issue statement, as a leakage control. One system is deliberately absent: a frontier model would be the natural comparison, but the only one available here also produced the labels, and a system evaluated against its own annotations measures nothing.

7.2 The pooled lexical score is an artefact of our sampling

The frozen dictionary reaches macro-F_1 of 0.946 on the random test set, which is not a difficulty measure. Every window is centred on the anchor, so a trigger-bearing item contains its trigger in every window and a control item contains none. The rule reduces to "is this item from the trigger family": its score is fixed by the stratum quotas and cannot vary with context width.

7.3 Supervised systems beat the dictionary where it was built to fail

On H1, built so the expression is present but is not this court announcing its own non-resolution, the rule reaches 0.895 while the sparse model reaches 0.947; it also edges the rule on PLAIN (0.962 against 0.942). The dictionary's pooled advantage comes entirely from the control strata, where it is correct by construction. The encoders show the pattern more strongly: the best reaches 0.891 on the random test set, and on the court-held-out condition legal-bert reaches 0.975 against the rule's 0.917.

It appeared as the development split grew, so we measured it directly. Figure 5 resamples training subsets of increasing size and evaluates each on the held-out H1 items. Accuracy rises monotonically, the spread across resamples narrows to nothing, and the share of subsets beating the dictionary goes from none at 45 items to all of them at 120, crossing over near 80. That is the shape of a learning curve rather than of noise, and it says the benchmark is nowhere near saturated.

7.4 Context helps up to a point and then hurts

For a window w we estimate, against the anchor sentence w_0,

$$\Delta(w) = F_1(y, \hat{y}_w) - F_1(y, \hat{y}_{w_0}), \qquad (3)$$

resampling the same items for both conditions, so the comparison is paired and isolates the window. Table 11 reports it. Widening to ± 256 characters costs nothing measurable on the random test

model	context	random (n=54)	temporal (n=14)	court (n=27)
Phi-3.5-mini-instruct	SENT	0.742	0.750	0.772
Phi-3.5-mini-instruct	W1024	0.742	0.713	0.760
Phi-3.5-mini-instruct	W256	0.774	0.713	0.760
Phi-3.5-mini-instruct	W4096	0.437	0.278	0.469
distilroberta-base	SENT	0.891	0.899	0.949
distilroberta-base	W1024	0.474	0.400†	0.470
distilroberta-base	W256	0.885	0.847	0.878
distilroberta-base	W4096	0.568	0.453	0.617
issue-only	ISSUE	0.541	0.630	0.692
legal-bert-base-uncased	SENT	0.884	0.869	0.975
legal-bert-base-uncased	W1024	0.645	0.602	0.596
legal-bert-base-uncased	W256	0.357	0.338	0.246
legal-bert-base-uncased	W4096	0.645	0.724	0.635
lexical-rule	any	0.946	0.862	0.949
majority	-	0.392†	0.400†	0.403†
tfidf-lr	SENT	0.813	0.792	0.897
tfidf-lr	W1024	0.567	0.621	0.681
tfidf-lr	W256	0.799	0.653	0.775
tfidf-lr	W4096	0.513	0.557	0.599

Table 3: Macro-F_1 on the three held-out conditions. The lexical rule is the frozen dictionary, invariant to the window because every window is anchor-centred, and issue-only is the leakage control. Splits are small and the temporal split too small to rank models. † marks a cell equal to the majority baseline, meaning the model collapsed to one class.

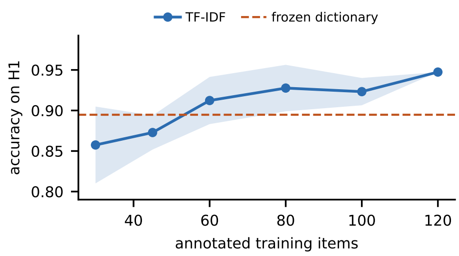

Figure 5: Learning curve on the adversarial stratum. Each point resamples twelve training subsets; bars are one standard deviation.

set (-0.014, 95% CI $[-0.110, 0.076]$). Widening further is costly and the effect is clear: ±1024 costs -0.245 and ±4096 costs -0.300 (95% CI $[-0.421, -0.167]$). The zero-shot model traces the same curve, from 0.742 at the sentence to 0.437 at its widest window.

The reading we prefer is that the disposition of one proposition is marked locally, and that a window wide enough to hold a second issue introduces its disposition as noise. That is a statement about fixed-window models rather than about context in principle. A system that locates the later holding sentence rule L1-3 turns on, rather than consuming everything around it, is left to future work.

7.5 Three checks

The issue-only classifier reaches 0.541 macro-F_1 against a majority baseline of 0.392, intervals barely overlapping. Every dictionary expression and holding cue is stripped from a statement before it is kept, so this is not lexical leakage but subject matter: which questions courts decline to reach correlates with what they are about. Part of what looks like reasoning is available from the topic alone.

Re-annotating 31 items with only the anchor sentence, the label matches the full-context label 90.3% of the time, and every flip falls in H1 and CTRL_HOLD. What changes most is the willingness to answer, abstention rising from 12.9% to 22.6%. For a person as for a model, context buys confidence more than accuracy.

Courts also label each other. A parenthetical reading "(declining to decide whether X)" asserts that the cited case did not resolve X, and we harvest 79,974 such characterizations. They serve as a resource rather than as validation, since if a later court says *Smith* held X and we say *Smith* left X open, either our label is wrong or the later court has overstated *Smith*, and the disagreement is consistent with both. Appendix M works through the cases that separate the systems, selected by a fixed rule rather than by hand.

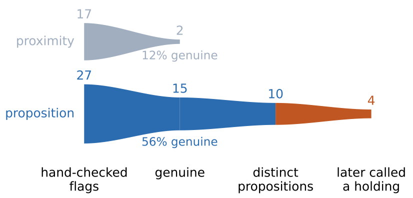

Figure 6: Attrition under the two matching strategies. Ribbon thickness is the count at each stage, so the first taper is the verified precision. The orange tail is the case series.

8 What Happens Next to an Unresolved Issue

Annotating citation chains one at a time does not scale. Of citing passages that survive a content-word filter, 78% are still about something other than the issue, so a usable observation costs roughly ten annotations. Instead we measure the phenomenon the way §5 measures prevalence, with frozen dictionaries applied to the whole corpus.

8.1 Matching propositions, not positions

Asking whether a non-resolution expression sits near the characterized passage is cheap but wrong: of 17 flagged cases only 2 were genuine, the expression usually governing a neighbouring question (Appendix L). Comparing propositions works. Writing $C(x)$ for the content words of a text, we match the origin's issue statement q against the later court's characterization c when

$$J(q,c) = \frac{|C(q) \cap C(c)|}{|C(q) \cup C(c)|} \geq \tau, \qquad \tau = 0.34, \tag{4}$$

so both sides of the comparison are propositions rather than positions in a document. Of 79,974 characterizations only 27 clear the threshold, few enough to hand-verify each: 15 are genuine, 56% precision against 12%, over 10 propositions.

8.2 What the verified cases show

Among those 10 propositions, each one a question the originating court demonstrably declined to decide, 4 are described by a later court as something that court held. The majority are reported accurately, with the later court repeating the reservation, as when one opinion notes that *Davis* reserved the question whether a request for counsel can be rendered ambiguous by preceding events.

Writing $S(o_1, q) \in \{L_0, \ldots, L_3\}$ for the status a later opinion o_1 ascribes to proposition q and $S_0(q)$ for what the origin did, the quantity of interest is

$$\Pr[S(o_1, q) = L_3 \mid S_0(q) = \textsc{unresolved}], \tag{5}$$

the chance that a question a court declined to decide is later called its holding. This is a case series and not a rate. Ten propositions cannot support an interval, and the threshold selects for verbatim quotation and so for the cases most likely to be characterized accurately, a bias against the phenomenon. It establishes that escalation occurs and that issue matching, not data, is the constraint. Appendix L elaborates.

9 Conclusion

Across 928,254 eligible federal appellate opinions, 18.92% contain at least one of thirteen frozen expressions of non-resolution. Declining to decide is not marginal. It is routine, announced in a vocabulary that has held its shape for a century, and rising in every era we can measure.

Pooled, the dictionary looks unbeatable, but that is fixed by the sampling. Per stratum the systems separate: on H1 the sparse model reaches 0.947 against 0.895, and on the court-held-out condition an encoder beats it outright. Context helps to ±256 characters and then degrades. The task is therefore not phrase recognition, and more text without a way to know what to look for in it makes systems worse.

For the empirical study of courts, the consequence is that a citation is a coarser instrument than it is usually taken to be. Network accounts of precedent read an edge as evidence of the cited opinion's authority (Fowler et al., 2007; Black and Spriggs, 2013), but an edge ascribing a holding to an opinion that expressly declined to reach the question means something different from one that follows a real holding, and the edge does not distinguish them. In 4 of 10 hand-verified propositions a later court described as settled what the originating court had reserved. Ten cases are a case series and not a rate, but they show the failure is real, and an instrument of this kind is what makes it countable at all.

The same gap runs downstream, since retrieval and summarization over case law inherit the assumption that a proposition attached to a citation was decided where it was cited. Separating disposition from discussion is a prerequisite for legal information systems, not an application of them.

Limitations

Who produced the labels

A single annotator applied the frozen guideline, so the labels carry no inter-annotator agreement statistic. A blind second pass over already-labelled items, shown in a different order with the first-pass labels withheld, reproduced them exactly (27 of 27, $\kappa = 1.00$). We report the figure and rest little on it, since perfect self-agreement is as consistent with an annotator recognising passages seen earlier as with a guideline that admits one reading. The quantity we do rest on is the within-annotator context ablation (§7), which measures how far the same annotator's decisions depend on material outside the anchor sentence. Whether two trained lawyers would agree on these labels is a question for a follow-up study with a practitioner panel.

The annotator is a language model, which places it in the same family as some of the systems under evaluation. Two design choices contain the circularity. The supervised baselines are trained from the labels but are architecturally independent of the label source, and they carry the context-scaling result. The zero-shot baseline is a small open model that had no part in producing the labels, saw no guideline, and was scored by label-token likelihood, so it is independent of them. A frontier model evaluated against its own annotations would report agreement with itself, so we omit that comparison. The escalation results in §8 rest on the same labels.

Benchmark and split sizes

Annotation produced 306 usable labels, 120 for development and 104, 36, and 46 for the three held-out conditions. At that size, differences of a few macro-F_1 points sit inside the noise, and the temporal condition carries wide intervals. The two comparisons we rest on are the per-stratum contrast, where the gap between the sparse model and the dictionary on H1 is large, and the paired context comparison, which is within-item and therefore better powered than the split sizes suggest. One fact a reader should weigh is that the separation between supervised systems and the dictionary emerged as the development split grew, having been absent at half this annotation scale. That is the signature of an effect that needed data to surface, and it is also a reason to rerun the benchmark at several thousand items.

The leakage control reaches 0.541 macro-F_1 against a majority baseline of 0.392, with intervals that barely overlap. Every dictionary expression and holding cue is stripped from a statement before it is kept, so the signal here is subject matter rather than lexical leakage of the label. Which questions courts decline to reach is correlated with what the questions are about, so part of what a system appears to infer from the passage is available from the topic alone. Future versions of this benchmark should either match strata on subject matter or report the issue-only control alongside every headline number, as we do here.

What the dictionary measures

A court can leave an issue open in ordinary prose. Freezing the thirteen expressions before annotation protects against tuning them to a result, and the recall audit in Appendix G bounds what they miss by reading a random sample of opinions containing none of them. Prevalence figures should be read as the rate of explicitly marked non-resolution. The item frame is scoped the same way. It admits anchors with a recoverable issue statement, and yield varies across expressions and, less strongly, across courts, so the benchmark samples anchors with a recoverable proposition. Appendix D reports yield by stratum so the direction of the loss is visible.

Corpus coverage

CourtListener is broad but uneven. Coverage, metadata completeness, and text quality vary by court and era, and the same opinion is sometimes stored more than once from different upstream sources. We deduplicate and report the eligibility funnel, and a change in measured prevalence across periods remains partly a change in what was digitized and how. This is why period-stratified rates appear throughout rather than a single time series presented as a trend in judicial behaviour.

A citing passage is treated as being about the origin's issue when its content words overlap the issue statement. That admits false positives, where a later opinion cites the origin for a neighbouring point sharing vocabulary, and false negatives, where a later court restates the issue in different words. Annotation removes the false positives, since off-issue passages are labelled and dropped, and the false-negative rate is unmeasured.

Scope of the comparison

The comparison between unresolved and decided issues matches on court, period, opinion length, and citation volume, and is descriptive. Whether an issue was left open is a judicial choice correlated with unobservables, including how hard and how contested the issue was, and those plausibly affect how later courts describe it. We report the contrast as an association.

Chains require the origin to carry a reporter citation locatable in the citing text, and are restricted to citing opinions inside the population, so district and state court treatments fall outside the sample and origins in the most recent years have had little time to accumulate citations. Both restrictions bias the chain sample toward older, better-reported, more-cited opinions.

The encoder baselines truncate at 512 tokens and the zero-shot model at 1600, which caps the widest context condition, so the comparison across context widths is partly a comparison of how much of the window each architecture uses. The zero-shot model has 3.8B parameters, and its role here is to show that an independent model responds to the context manipulation in the same direction. The study covers English-language United States federal appellate opinions, and extension to state courts, trial courts, agency adjudication, or other legal systems remains future work.

Ethics Statement

The corpus consists of published judicial opinions, which are public records, and we add no information about any individual beyond what those opinions already contain. Opinions name parties, including criminal defendants, immigration petitioners, and civil litigants. We release passage offsets and opinion identifiers rather than redistributing text, so that removals and corrections upstream propagate. The task is descriptive. A system that classifies issue disposition should not be used to characterize a court's work without review, and the escalation analysis measures how propositions are described in later opinions, not whether any court erred. We name no judge in any reported result.

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A Excluded Federal Courts

CourtListener assigns federal jurisdiction codes to many bodies beyond the fourteen courts in the population. Every excluded court is listed in the released artefact `01_courts.json`. They fall into four groups: the historical circuit courts that sat before the Judiciary Act of 1891, appearing under roughly 130 identifiers of the form `circt*`; abolished or specialized Article I and Article III courts, among them the Court of Customs and Patent Appeals, the Court of Claims, and the Court of International Trade; the Foreign Intelligence Surveillance Court and its Court of Review; and administrative review bodies whose decisions are published alongside judicial opinions, among them the Board of Immigration Appeals, the Merit Systems Protection Board, and the Patent Trial and Appeal Board.

Excluding administrative bodies is a substantive choice. Agency adjudicators use the same vocabulary of non-resolution, but they are not courts of appeals, their decisions carry different precedential force, and the citation practices layer two depends on differ. The historical circuit courts are excluded for a different reason: their reporting is uneven and their text largely OCR-derived, so a rate computed over them would measure digitization as much as judicial practice.

B Text Extraction

CourtListener stores an opinion's text in whichever markup flavours its upstream sources supplied. We take the first field with usable content in the order `html_columbia`, `html_lawbox`, `xml_harvard`, `html_anon_2020`, `html`, `plain_text`, `html_with_citations`. The citation-annotated variant is last because it injects anchor text into the body. Markup is removed by closing block-level tags into newlines, deleting remaining tags, unescaping entities, and normalizing whitespace; character offsets in the released benchmark refer to this normalized text. Table 4 reports which field supplied the text, by period. Source composition shifts with period because different digitization projects covered different eras, which is one reason we report period-stratified rates rather than a single pooled trend.

C The Frozen Trigger Dictionary

Figure 3 lists the thirteen expressions with their corpus frequencies. The regular expressions themselves are reproduced below; they were fixed before annotation and their SHA-1 is `c23210667a4e`.

T08 and T09 are proper subsets of T07 by construction. They are retained because the design names them separately, and a passage matching several expressions is counted once in opinion-level rates while keeping every identifier. T11 is the only expression not matched as written, for the reason given in §5.

D Issue Statement Extraction

An item needs a statement of the proposition at stake that does not reveal the answer. We derive it from the sentence containing the anchor rather than from a fixed character budget, which stops the clause running into whatever the court said next.

The procedure takes the material following the anchor within the anchor's sentence, deletes reporter citations and signals, and looks for a *whether*-complement within the first sixty characters. If none is present but the clause opens with a complementizer or a preposition taking a propositional object, it is rewritten as a *whether*-clause. If it opens with neither, it is kept as a noun phrase, but only when its first token is not a verb, negation, conjunction, or pronoun, since those signal a fragment rather than a statement of an issue. Where the trigger closes its own sentence, the material before the anchor is used instead, and only when that material itself names the issue with *whether*, *the question of*, or *the issue of*. Two filters run last. Any dictionary expression or holding cue in the candidate statement is cut, which is what allows the claim that no issue statement contains one, and statements with fewer than three content words, or more than forty percent non-alphabetic, are rejected as extraction debris.

Rejection is not random. An anchor with no extractable issue statement does not enter the frame, so the benchmark is a sample of anchors with a recoverable proposition rather than of all anchors. Table 5 reports the yield by expression and by court so that the direction of any bias is visible.

E Annotation Guidelines

The guidelines below are reproduced verbatim from the frozen protocol. Version 1.0 was drafted before the pilot; version 1.1 incorporates the single revision the design permits, made once after the pilot and before any main-sample annotation.

```
TASK. You are given a legal issue and a passage from
  a United States federal
appellate opinion. Decide whether the opinion
  resolved that issue.

LABELS.
  DECIDED     The authoring court resolved the
    issue. It stated a holding, a
              conclusion, or a
                determination on the
                issue, whether or not
                that
              resolution was necessary
                to the judgment, and
                whether or not it
              was later characterized
                as dictum.
  UNRESOLVED  The authoring court did not resolve
    the issue for itself.

If and only if the label is UNRESOLVED, assign one
  sublabel.
  AVOIDED     The court declined to reach the
    issue because some other ground
              disposed of the case or
                the claim. Typical
                form: "because we
              affirm on qualified
                immunity grounds, we
                need not decide
                whether
              the search was lawful."
  ASSUMED     The court proceeded on a stated
    assumption about the issue and
              resolved the case on
                other grounds, so that
                the assumption did
              no independent work.
                Typical form:
                "assuming without
                deciding
              that the statute reaches
                this conduct, the
                evidence is
              insufficient."
  RESERVED    The court expressly held the issue
    open for future resolution,
              signalling that it
                remains available.
                Typical form: "we
                leave for
              another day whether
                Chevron survives in
                this context."

DECISION RULES.
  L1-1  ATTRIBUTION. The label describes what THIS
    opinion did. Language about
        what a different court did, whether
          a lower court, a sister circuit,
          or
        the Supreme Court, does not by
          itself make this opinion's
          disposition
```

field	pre-1990	1990-2004	2005-2014	2015-2026
`html`	17.2	30.1	2.3	0.0
`html_lawbox`	10.5	0.6	12.9	0.0
`plain_text`	0.0	13.3	16.8	67.7
`xml_harvard`	72.3	56.0	68.0	32.3

Table 4: Text source field by period, as a percentage of eligible opinions in that period.

```
T01  need(?:s|ed)?\s+not\s+(?:\w+\s+){0,2}?decide\b
T02  need(?:s|ed)?\s+not\s+(?:\w+\s+){0,2}?reach\b
T03  \bdo(?:es)?\s+not\s+(?:\w+\s+){0,2}?decide\b
T04  \bdo(?:es)?\s+not\s+(?:\w+\s+){0,2}?reach\b
T05  declin(?:e|es|ed|ing)\s+to\s+(?:\w+\s+){0,2}?decide\b
T06  declin(?:e|es|ed|ing)\s+to\s+(?:\w+\s+){0,2}?address\b
T07  \bwithout\s+deciding\b
T08  \bassuming\s+(?:\w+\s+){0,4}?without\s+deciding\b
T09  assum(?:e|es|ed)\s+(?:\w+\s+){0,4}?without\s+deciding\b
T10  express(?:es|ed|ing)?\s+no\s+(?:\w+\s+){0,2}?opinion\b
T11  reserv(?:e|es|ed|ing)\s+(?:\w+\s+){0,3}?(?:question|questions|issue|issues|matter|judgme
    nt|decision)\b|reserv(?:e|es|ed|ing)\s+(?:\w+\s+){0,3}?for\s+another\s+(?:day|case)\b
T12  leav(?:e|es|ing)\s+(?:\w+\s+){0,4}?(?:for|to|until)\s+another\s+(?:day|case|occasion)\b
T13  \b(?:not|un)\s*necessary\s+to\s+(?:decide|reach|resolve|address)\b
```

Figure 7: The thirteen frozen expressions as regular expressions, matched case-insensitively over whitespace-normalized text.

id	anchors	with issue	yield (%)
T01	49,640	34,951	70.4
T02	30,909	16,737	54.1
T03	22,309	15,820	70.9
T04	45,401	25,678	56.6
T05	8,144	5,871	72.1
T06	21,004	10,165	48.4
T07	30,894	23,189	75.1
T08	4,718	3,420	72.5
T09	7,779	5,663	72.8
T10	24,641	17,710	71.9
T11	34,080	15,684	46.0
T12	3,508	1,826	52.1
T13	18,876	11,670	61.8

Table 5: Issue-statement extraction yield by expression.

```
        UNRESOLVED. "The district court
          concluded it need not decide X"
          tells
        you about the district court. Read
          on to see what this court did.
L1-2  UNIT. In a cluster containing a majority
    opinion and separate opinions,
        each opinion is its own unit. A
          dissent that would decide an issue
          the
        majority avoided is DECIDED for the
          dissent and UNRESOLVED for the
        majority.
L1-3  LATER RESOLUTION CONTROLS. A court may
    write "we need not decide X" and
        then decide X anyway, in the same
          opinion, in an alternative
          holding, in
        a footnote, or in a later section.
          Where the opinion does resolve the
        issue somewhere, the label is
          DECIDED. A court's
          characterization of
        what it need not do controls only
          where no independent resolution
        follows.
L1-4  ASSUMPTION THAT DOES WORK. If the court
    assumes a proposition and the
        assumption is then treated as
          settled for the remainder of the
          analysis
        in a way that determines the
          outcome, the label remains
          UNRESOLVED and
        the sublabel is ASSUMED. Assuming a
          proposition is not deciding it,
          even
        when the assumption drives the
          result.
L1-5  NO TRIGGER REQUIRED. UNRESOLVED does not
    require any particular phrase.
        A court can leave an issue open in
          ordinary prose.
L1-6  PARTIAL RESOLUTION. The unit is the
    proposition named in the issue
        statement, not the broader legal
          question it belongs to. If the
          court
        resolves the anchored proposition
          but leaves a neighbouring one
          open,
        the label is DECIDED. If the court
          resolves a neighbouring
          proposition
        but leaves the anchored one open,
          the label is UNRESOLVED.
L1-7  REMANDS. An issue remanded for the
    district court to address in the
        first instance is UNRESOLVED,
          sublabel AVOIDED, unless the
          appellate
        court also states the governing rule
          and applies it.
L1-8  ABSTAIN. If the passage is too corrupt,
    too truncated, or too ambiguous
        to support either label, output
          UNCLEAR rather than guessing.
L1-9  ISSUES THIS OPINION NEVER TAKES UP.
    Sometimes the only reason an issue
        is in front of you is that the
          opinion mentioned, in a
          parenthetical or
        in a description of some other
          court's work, that a different
          court did
        not decide it, and this opinion
          never reaches the issue in its own
        right. That is neither DECIDED nor
          UNRESOLVED for this opinion.
          Output
        UNCLEAR. This rule does not apply
          where the opinion goes on to
          engage
        with the issue itself: there, L1-1
```

```
and L1-3 govern.
```

Version 1.1. The single permitted revision after the pilot, and the layer-two and issue-statement guidelines, are in the released package.

F Pilot Study and Protocol Revision

The pilot ran on a preliminary frame built from a leading portion of the bulk opinion table, before the full pass was committed to. Its purpose was to find out whether the guideline could be applied at all, whether the controls behaved, and which strata were worth sampling heavily. Sixty items were annotated. Pilot items are not part of the benchmark and no pilot label appears in any result.

F.1 What the pilot showed

Labels fell out by stratum as follows: CTRL_HOLD 12 DECIDED and 0 UNRESOLVED; PLAIN 19 UNRESOLVED and 1 DECIDED; H2 8 UNRESOLVED and 2 DECIDED; H1 5 UNRESOLVED, 5 DECIDED, and 5 UNCLEAR.

Three things follow. The holding-cue control behaves as intended and is not contaminated. The PLAIN stratum is nearly degenerate: a trigger with no structural complication almost always does mean the court left the issue open, which is why a keyword rule scores as well as it does and why that score means little. The H1 stratum is close to an even split, so it is where the task actually lives, and the sample was reallocated toward it. This reallocation changes only how many items are drawn per stratum. Label definitions and stratum definitions were fixed before the pilot and held throughout.

F.2 The one protocol revision

Version 1.0 of the guideline had no rule for a passage whose only trigger sits inside a parenthetical or a description of another court's work, on a question the present opinion never reaches in its own right. A citation noting that the Supreme Court declined to decide whether an airport user fee was excessive tells you nothing about the disposition of the Federal Circuit opinion that cites it, and forcing a choice between DECIDED and UNRESOLVED would have manufactured a label. Rule L1-9 routes these to UNCLEAR. It uses an existing label; it adds none. This is the single revision the design permits, it was made once, before any main-sample annotation, and the changelog is reproduced in Appendix E.

F.3 Three pipeline defects the pilot exposed

These concern candidate construction, not the guideline. The pattern flagging a trigger as plausibly describing another institution was case-sensitive, matching "the district court" but missing "the District Court," which appellate opinions write about as often; it now also recognizes "the court held" and its near neighbours. The annotator was shown a later holding sentence only for items already flagged H2, though Rule L1-3 turns on precisely that passage, so an H1 item whose opinion later resolved the issue was unanswerable rather than hard; the sentence is now shown whenever one exists outside the displayed window. And "our holding" was a control locator, but in the pilot it pointed at a named prior case rather than the present disposition every time it fired, so it was dropped.

F.4 What the pilot did not do

It was not used to adjust any label boundary. The trigger dictionary was frozen before the pilot and is unchanged; in particular, the pilot surfaced two recurring dictionary false positives, and neither was patched. One is the statutory-scope reading of *reach*, as in "the shipment clause does not reach shipments between the continental States," which is a holding about a statute rather than a non-resolution. The other is a line-break artefact in the source text that splits *preserving* into `P` and `reserving`, which then matches T11. Both stay in, are labelled DECIDED by the guideline, and are counted against the dictionary in §7.

G Dictionary Recall Audit

The frozen dictionary bounds what candidate generation can find. It says nothing about non-resolution expressed in language outside it. To bound that, we drew a random sample of eligible opinions containing no dictionary expression and read them for issues the court discussed and left unresolved.

Reading whole opinions at random is a weak audit. A non-resolution can sit anywhere in forty pages, most opinions contain none, and an annotator who has internalized the dictionary is primed to see the vocabulary it already covers.

We use a second, deliberately wider probe list instead. Fifteen expressions outside the frozen dictionary were written after the fact and run over a random sample of 4,000 opinions the dictionary called clean. The probe is confined to this audit and enters neither candidate generation, the benchmark, nor any prevalence figure.

17.8% of clean opinions contain at least one probe expression. Table 6 gives the breakdown. What that figure bounds needs stating carefully. It is a measure of *lexical* coverage, not of missed non-resolution. Some probe hits are genuine, as in a Seventh Circuit opinion that writes "Volkswagen does not raise this argument and we do not resolve it," and then says the question "remains unsettled." Others are the same statutory-scope reading of *reach* and *address* that already contaminates the frozen dictionary, where the court is saying what a statute covers rather than what it declined to decide. Annotating a sample of probe hits would be needed to convert the probe rate into an estimate of missed non-resolution, so we report it as a coverage figure.

The largest omission is the *address*, *consider*, *resolve* family. The frozen dictionary contains *need not decide* and *need not reach* but not *need not address*, and *decline to address* but not *decline to consider*. The design named thirteen expressions and they were frozen before annotation, so the gap stands rather than being patched. The pilot had already flagged one instance, in a control opinion reading "we need not address the government's contention" that was nonetheless admitted to the zero-trigger pool.

Two consequences follow for how the paper's numbers should be read. Population prevalence in §5 measures explicitly marked non-resolution using one fixed vocabulary, and a wider vocabulary would find more, so it is a floor rather than an estimate of the phenomenon. And the control pool is contaminated in a known direction, since some opinions admitted as trigger-free do contain unresolved issues expressed in language outside the dictionary. That contamination makes the controls harder rather than easier, so it works against every reported separation rather than for it.

H Full Prevalence Tables

I Cross-Check Against the Search Index

The bulk release and the public CourtListener search index are built from the same database but refreshed on different schedules and tokenized differently. Querying the index for the thirteen literal phrases, restricted to the fourteen courts, gives an estimate of how many opinions contain each expression that is independent of our text extraction,

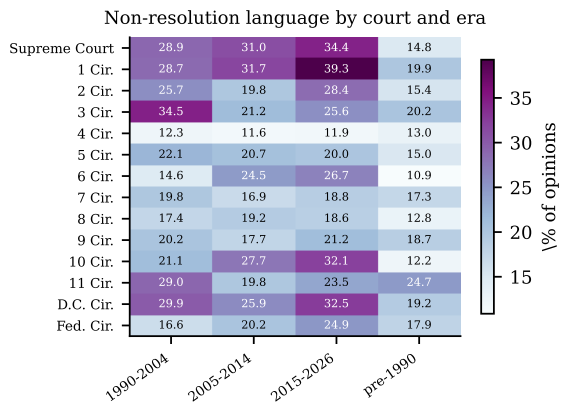

Figure 8: Share of eligible opinions containing at least one frozen dictionary expression, by court and period. Cells with fewer than 150 eligible opinions are left blank rather than plotted as noise.

markup handling, and field precedence.

Three hundred count queries were issued against the v4 search endpoint: a denominator for each of the fourteen courts, a denominator for each court and period, and a count for each of the thirteen literal phrases, both per court and pooled. No opinion text was retrieved.

Two differences from the bulk pipeline are expected and neither is a defect. The index counts opinions as the search system stores them, while our denominator counts opinions surviving the eligibility criteria of §4, which remove short orders and duplicates; the index denominator is therefore the larger of the two. And the index matches literal phrases, while the frozen dictionary tolerates a few intervening words and constrains T11, so the two counts are not measuring the same event. The comparison is useful precisely because the two disagree in known directions: a large unexplained gap would indicate that markup handling or field precedence was losing text, and a close match on the shape of the distribution across courts and expressions indicates that it is not.

The pooled index count over the thirteen literal phrases restricted to the fourteen courts is 108,669 opinions against a denominator of 1,388,276 indexed opinions, an implied rate of 7.8%. Ranking the expressions by frequency, the index and the bulk corpus agree on the ordering of the top group, with *need not decide* first in both.

J Supplementary Tables

These are collected rather than placed inline because each is a secondary cut on a quantity already

id	probe expression	n	% of clean
P01	need not address	128	3.20
P02	need not consider	92	2.30
P03	need not resolve	46	1.15
P04	no occasion to decide	14	0.35
P05	leave open the question	45	1.12
P06	decline to consider/resolve/reach	80	2.00
P07	assume arguendo	111	2.77
P08	we may assume	44	1.10
P09	pretermit	18	0.45
P10	do not address/consider/resolve	254	6.35
P12	for purposes of this appeal we assume	1	0.03
P13	intimate no view	9	0.22

Table 6: Probe expressions found in opinions the frozen dictionary called clean. The probe was written after the fact and is used only for this audit.

court	1990-2004	2005-2014	2015-2026	pre-1990
Supreme Court	28.88	30.96	34.35	14.78
1 Cir.	28.67	31.66	39.27	19.87
2 Cir.	25.66	19.77	28.40	15.42
3 Cir.	34.48	21.23	25.60	20.21
4 Cir.	12.30	11.62	11.85	13.00
5 Cir.	22.07	20.69	20.02	15.01
6 Cir.	14.63	24.52	26.73	10.87
7 Cir.	19.80	16.86	18.77	17.33
8 Cir.	17.43	19.17	18.60	12.82
9 Cir.	20.24	17.72	21.17	18.68
10 Cir.	21.11	27.65	32.08	12.23
11 Cir.	28.97	19.78	23.54	24.72
D.C. Cir.	29.92	25.94	32.50	19.18
Fed. Cir.	16.59	20.22	24.93	17.89

Table 7: Share of opinions containing at least one dictionary expression, by court and period.

type group	N	k	%
majority/lead	864,804	169,373	19.59
other	1,664	101	6.07
separate	61,786	6,136	9.93

Table 8: By opinion type.

precedential status	N	k	%
Errata	104	3	2.88
Published	767,586	160,008	20.85
Relating-to	552	21	3.80
Unknown	3,291	1,139	34.61
Unpublished	156,706	14,439	9.21

Table 9: By precedential status.

reported.

K Model Details

K.1 Context conditions

Every system is evaluated under four windows around the anchor: the anchor's sentence (SENT), and ± 256, ± 1024, and ± 4096 characters. Windows are cut from the normalized text and whitespace-collapsed. The sentence condition isolates local lexical evidence; the widest condition is the only one that routinely reaches a later holding sentence in the same opinion, which is what the H2 stratum requires.

K.2 Lexical rule

The floor. The frozen dictionary is applied to the context window and the item is predicted UNRESOLVED when any expression matches. This is not a strawman: it is exactly the procedure that generated the candidates, and it is what a keyword system would do. Because controls are drawn from opinions containing no dictionary expression anywhere, the rule is close to perfect on the CTRL strata by construction, and its overall number is therefore an optimistic reading of what phrase matching achieves. The strata that matter are H1 and H2, where the rule is wrong whenever the annotation says the court's actual disposition runs against the phrase.

court	indexed opinions
Supreme Court	498,145
1 Cir.	37,036
2 Cir.	88,338
3 Cir.	65,415
4 Cir.	66,616
5 Cir.	117,894
6 Cir.	62,972
7 Cir.	71,787
8 Cir.	79,172
9 Cir.	137,966
10 Cir.	40,431
11 Cir.	53,966
D.C. Cir.	40,448
Fed. Cir.	28,090

Table 10: Denominators reported by the CourtListener search index, used only for the cross-check in Appendix I.

split	context	ΔF_1	95% CI	$P(\Delta>0)$
TEST	W256	-0.014	[-0.110, +0.076]	0.363
TEST	W1024	-0.245	[-0.365, -0.127]	0.000
TEST	W4096	-0.300	[-0.421, -0.167]	0.000
TEMPORAL	W256	-0.139	[-0.315, +0.017]	0.033
TEMPORAL	W1024	-0.171	[-0.409, +0.048]	0.057
TEMPORAL	W4096	-0.236	[-0.462, +0.026]	0.037
COURT	W256	-0.123	[-0.299, +0.026]	0.042
COURT	W1024	-0.216	[-0.387, -0.055]	0.004
COURT	W4096	-0.298	[-0.482, -0.105]	0.003

Table 11: Paired bootstrap on the effect of widening the context window, relative to the anchor sentence alone, for the sparse model. Resamples are paired across conditions.

K.3 Sparse baseline

TF-IDF over word unigrams and bigrams, minimum document frequency two, at most 60,000 features, sublinear term frequency, followed by ℓ_2-regularized logistic regression with $C = 4$ and balanced class weights. Trained on DEV only.

K.4 Encoders

LEGAL-BERT (Chalkidis et al., 2020) and DistilRoBERTa, fine-tuned for five epochs at learning rate 2×10^{-5}, batch size 16, maximum length 512 tokens, with the issue statement as the first segment and the context window as the second, truncating only the context. Class weights are inverse to DEV frequency. The 512-token cap means the ± 4096 condition is truncated for these models, so their context curve understates what a long-context architecture could use; this is stated in the Limitations.

K.5 Leakage control

A classifier trained on the issue statement alone, with no context at all. Issue statements are constructed to be neutral and any dictionary expression or holding cue is cut from them, so this classifier should sit near the majority-class baseline. It is the check on how much of the benchmark is answerable from the phrasing of the issue alone.

K.6 Uncertainty

Macro-F_1 intervals are percentile bootstrap over 2,000 resamples of the evaluation set. Each item comes from a distinct opinion, so resampling items and resampling opinions coincide here; the clustered bootstrap is needed only in §8, where several citing passages can share an origin.

K.7 Splits

The four evaluation conditions are disjoint and nested in a fixed order. Every item from the two reserved courts forms COURT. Of what remains, every item filed in 2018 or later forms TEMPORAL. The remainder is split at random into TEST and DEV, and DEV is the only portion any system was fit on. The hard-negative subset is not a split: it cuts across all of them, and is reported as a breakdown.

K.8 Zero-shot language model

Phi-3.5-mini-instruct (revision `2fe192450127`) runs on one Tesla T4 with unquantized float16 weights. Logits are cast to float32 before scoring. Nothing is generated: for each item we form one prompt and compare the summed log-probability of the continuation `DECIDED` against `UNRESOLVED`. Scoring rather than generating removes refusals, verbose answers needing parsing, and format drift that varies with prompt length, all of which would otherwise be confounded with difficulty. Prompts truncate to 1600 tokens, which binds only in the widest condition. The prompt is fixed across conditions and reads, in full: a one-line framing that the passage is from a United States federal appellate opinion, the issue, the pas-

period	N	k	%	95% CI	per 100k words
pre-1990	368,842	58,577	15.88	15.76–16.00	7.9
1990-2004	257,715	54,071	20.98	20.82–21.14	10.1
2005-2014	180,721	35,908	19.87	19.69–20.05	10.3
2015-2026	120,976	27,054	22.36	22.13–22.60	10.7

Table 12: The same quantity by era. The rate per hundred thousand words is reported alongside the opinion-level rate because opinion length is not constant across eras.

split	n	% unresolved
COURT	54	57.4
DEV	143	53.8
TEMPORAL	44	54.5
TEST	123	54.5

Table 13: Evaluation splits. They are disjoint at the opinion level and only DEV was used for fitting.

stratum	n	agreement with full context (%)
CTRL_HOLD	4	75.0
H1	11	81.8
H2	7	100.0
PLAIN	9	100.0

Table 14: How often the annotator's sentence-only label matches the label the same annotator gave with the full window. A within-annotator context ablation, not an inter-annotator statistic.

stratum	context	n	lexical	TF-IDF
CTRL_HOLD	SENT	44	1.000	0.750
CTRL_HOLD	W1024	44	1.000	0.318
CTRL_HOLD	W256	44	1.000	0.545
CTRL_HOLD	W4096	44	1.000	0.227
CTRL_RAND	SENT	9	1.000	0.222
CTRL_RAND	W1024	9	1.000	0.556
CTRL_RAND	W256	9	1.000	0.444
CTRL_RAND	W4096	9	1.000	0.333
H1	SENT	38	0.895	0.947
H1	W1024	38	0.895	0.816
H1	W256	38	0.895	0.947
H1	W4096	38	0.895	0.842
H2	SENT	43	0.907	0.907
H2	W1024	43	0.907	0.814
H2	W256	43	0.907	0.930
H2	W4096	43	0.907	0.814
PLAIN	SENT	52	0.942	0.962
PLAIN	W1024	52	0.942	0.885
PLAIN	W256	52	0.942	0.923
PLAIN	W4096	52	0.942	0.865

Table 15: Accuracy by stratum and context width. The lexical rule is the frozen dictionary applied to the window.

sage, and the question whether the court resolved that issue or left it unresolved, answerable in one word. No few-shot examples are supplied and the annotation guideline is not shown, so the model sees strictly less than the annotator did.

L Judicial Characterizations and Layer Two

One annotator produced the study's labels. This appendix reports a check built from a source no annotator touched, and the citation machinery behind §8.

L.1 Judges label each other's opinions

When a court writes "*Smith*, 123 F.3d 456, 460 (declining to decide whether the statute applies)," it asserts on the record that *Smith* did not resolve that question; "(holding that . . .)" asserts the opposite. We scanned all 928,254 eligible opinions for reporter citations and applied a frozen two-class characterization dictionary to a window around each. 4,863,357 passages matched, but the loose form is unreliable: a marker near a citation is often about something else, as where "it did not reach a verdict" refers to a jury. Requiring the verb to bind directly to the citation, as a parenthetical or appositive, leaves 79,974 passages, 1,077 of them asser-

context	stratum	accuracy (%)
SENT	CTRL_HOLD	86.4
SENT	CTRL_RAND	22.2
SENT	H1	84.2
SENT	H2	69.8
SENT	PLAIN	82.7
W1024	CTRL_HOLD	79.5
W1024	CTRL_RAND	66.7
W1024	H1	84.2
W1024	H2	72.1
W1024	PLAIN	73.1
W256	CTRL_HOLD	79.5
W256	CTRL_RAND	44.4
W256	H1	84.2
W256	H2	74.4
W256	PLAIN	84.6
W4096	CTRL_HOLD	88.6
W4096	CTRL_RAND	77.8
W4096	H1	28.9
W4096	H2	30.2
W4096	PLAIN	21.2

Table 16: Zero-shot open instruction-tuned model, accuracy by context width and sampling stratum, scored by label-token likelihood.

tions of non-resolution, over 22,471 cited opinions. We release this set.

L.2 Why it cannot validate the annotations

The obvious use fails. Suppose a later court says *Smith* held X and our annotation says *Smith* left X open. Two explanations fit: our label is wrong, or the later court has overstated *Smith*, which is the phenomenon this paper measures. Nothing in the disagreement distinguishes them. We report this because the design is tempting and the flaw shows only once the numbers are in front of you.

L.3 Proximity matching and why it failed

Pairing each characterization with the sentence in the cited opinion sharing the most content words gives 2,252 items (353 unresolved, 1,899 decided), but systems transfer poorly: the frozen dictionary reaches 0.567 macro-F_1 against 0.946 on the annotated benchmark. The cause is measurable and is not the labels. Among items a judge called unresolved, only 8.2% carry a dictionary expression in the selected anchor sentence, against 0.5% of items a judge called decided: enriched sixteenfold, so the labels carry signal, but in more than nine cases in ten the matcher selects where the opinion *discusses* the issue rather than where it *disposes* of it. Accuracy rises monotonically with match quality, from 0.485 to 0.610, the pattern anchoring noise predicts and label noise does not. Asking instead whether a dictionary expression sits within 400 characters of the characterized passage covers 58,600 characterizations, of which 2 of 17 hand-checked flags were genuine. Both failures are the same failure, and it is why §8 matches propositions.

L.4 Chains, matching, and inference

Absolute rates in §8 are not prevalence estimates, since the characterization set is dominated by holding-ascriptions by construction, so only contrasts are interpretable. The origin-side reading is high precision and low recall, so the comparison runs between origins that *visibly* mark an issue open and the rest. Intervals resample originating opinions rather than passages, since several citing passages share an origin. We fit no causal model: whether a court leaves an issue open is a choice correlated with unobservables, including how contested the question was, and those plausibly affect how later courts describe it.

Locating where an opinion disposes of a named proposition, rather than where it discusses one, is the binding constraint on a population rate. At the measured yields, estimating one to within a few points needs on the order of a thousand verified propositions, which at one in ten means annotating close to ten thousand citing passages: an annotation budget an order of magnitude larger than a single annotator supplies.

M Error Analysis

The cases that separate the systems are more informative than the aggregate numbers, so a fixed selection rule was applied rather than a hand-picked illustration. Three groups were extracted from the held-out splits: items the frozen dictionary gets wrong, items where widening the context window moved the zero-shot model from wrong to right, and items where widening it moved the model from right to wrong. The third group exists because a selection showing only context helping would be an advertisement rather than an analysis.

context broke it *(H1, ca9 2011; gold UNRESOLVED, dictionary says UNRESOLVED)*

ISSUE: whether Levi Strauss also had two factors– acquired distinctiveness and degree of recognition–that weighed in its favor

PASSAGE: . . . relevant factors convinces us that application of the incorrect standard affected its dilution determination. According to the district court, degree of similarity was only one of three factors that weighed in Abercrombie's favor. The district court assumed, [[without deciding]] , that Levi

Strauss also had two factors– acquired distinctiveness and degree of recognition–that weighed in its favor. Thus, application of ...

context fixed it *(CTRL_HOLD, ca2 1968; gold DECIDED, dictionary says DECIDED)*
ISSUE: whether Hammond would not be "in custody" even if he were on active military duty
PASSAGE: ... how the interests of justice would be served or the question before us would be meaningfully different had Hammond first reported for active duty and then applied for the writ. Indeed, we can only understand the government's position on the assumption – which [[we reject]] – that Hammond would not be "in custody" even if he were on active military duty. Nor do we discern any distinguishing features because Hammond or ...

dictionary misled *(H1, ca1 1992; gold DECIDED, dictionary says UNRESOLVED)*
ISSUE: for appeal; b) the Fund waived its challenge to venue; and c) the district of New Hampshire was the proper venue. 3
PASSAGE: ... e 10 In a prior order entered on April 30, 1991, the district court denied the Fund's motion to dismiss for improper venue. On appeal the Fund also challenges the propriety of that order. MKF on the other hand claims that a) the Fund failed to properly p [[reserve the issue]] for appeal; b) the Fund waived its challenge to venue; and c) the district of New Hampshire was the proper venue. 3 11 MKF filed its complaint ...

The dominant failure of the dictionary is the one the H1 stratum was built around. An opinion recounting the procedural history below reproduces the lower court's language, and a rule matching on the phrase attributes that court's restraint to the court now writing. Quotation is the other frequent source, where the opinion quotes a precedent that declined to decide something and then decides it. Where the wider window helps, it usually supplies the later holding sentence rule L1-3 turns on. Where it hurts, the added material introduces a second, neighbouring issue whose disposition differs from the anchored one, the failure mode rule L1-6 governs for annotators and for which the models have no comparable instruction.

N Secondary Prevalence Cuts

Opinion type and precedential status were named as secondary analyses before the corpus was built. Majority and lead opinions carry the expressions at 19.59% against 9.93% for concurrences and dissents, and published opinions run at 20.85% against 9.21% for unpublished dispositions. Appendix H gives both tables. Neither cut is offered as an explanation, and both are confounded with opinion length and with which courts publish what.

O Sublabel Distribution

Within UNRESOLVED, the sublabels split 86.9% AVOIDED, 12.6% ASSUMED, and 0.5% RESERVED. The imbalance is itself informative. Courts most often decline to reach an issue because something else disposes of the case, the narrow-grounds discipline; expressly holding a question open for a future case is much rarer, and assuming a proposition to show it does not matter sits in between.

P Reproduction and Compute

Bulk processing and the non-LLM stages run on one laptop with ten cores and 17 GB of memory. The replacement zero-shot evaluation runs on one Tesla T4 with float16 weights; scoring all twelve context-by-split cells takes 12.1 minutes after model loading.

P.1 Bulk processing

The five bulk tables total roughly 62 GB compressed. The opinion table alone is 54.6 GB and expands 7.0-fold to about 384 GB, which is never written to disk: decompression is delegated to a subprocess so it overlaps with parsing, and a single streaming pass filters to the population, strips markup, matches the dictionary, and writes sharded text with a byte-offset index. Decompression sustains about 71 MB/s and CSV parsing about 70 MB/s, so the pass is bounded by their pipeline rather than by anything downstream, and completes in 114 minutes. Later stages seek into the shards by offset and never reread the bulk file. Two gates keep the pass cheap. The population test is applied to the raw row before any markup is stripped, so rows outside the fourteen courts cost only CSV tokenization, and a substring check precedes the regular expression union, so the union is run only on documents that could match.

P.2 Models

The sparse baselines train in seconds and the encoders fine-tune on the Apple Silicon Metal backend in a few minutes per context condition. The zero-shot replacement is the only model evaluated on the Tesla T4; its larger weights do not fit the laptop setup used for the other stages.

P.3 Network

The CourtListener REST API is used only for the independent cross-check in Appendix I, which issues roughly three hundred count queries. Nothing in the main pipeline depends on per-record API access, which is the point of working from the bulk release: the citation graph in particular would otherwise require one request per edge.

P.4 Released artefacts

We release the frozen dictionary with its SHA-1, the eligibility funnel, the item frame, the benchmark with splits and labels, the issue-specific chains, all model predictions, and the pipeline. The benchmark is distributed as opinion identifiers with character offsets into the normalized text, together with the normalization code, so that upstream corrections and removals propagate.